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Appraisal of Mechanical and Tribological Properties on PA6-TiO₂ Composites through Fused Deposition Modelling

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Abstract

The primary objective of this examination is to build an arrangement of new polyamide blends with inventive properties like biodegradability. The composite blends of polyamide 6 (PA6) blended with 10, 20 and 30 wt.% titanium-di-oxide (TiO₂) were taken for this proposed work and converted in a form of wire in a filament extruder machine and taken to build the parts by using Fused deposition modelling (FDM) machine. Fused deposition modelling (FDM) is an additive manufacturing technology commonly used for modelling, prototyping, and production applications. This process constructs three-dimensional objects directly from 3D CAD data. A temperature-controlled head extrudes thermoplastic material layer by layer. The quality and strength of part that is built depends on many variables in FDM process. In this study important process parameters of extrusion were fixed. Test specimens were prepared by melting, compounding and smoothly printed by means of a commercial FDM three-dimensional (3D) printer. The printed samples were taken for mechanical and tribological test. The tested sample's outcome from UTM indicates that incorporation of filler stacking with equally distributed particles and higher weight (30) percentage of TiO₂ has influenced in higher mechanical properties of virgin PA6 and lower weight (10 and 20) percentage of TiO₂ gives moderate mechanical properties of the same. These tribological tests were carried out on the pin on disk tribometer under varying load conditions such as 5, 10, 15 and 20 N at sliding speed of 1.256 m/s for the run time of 5 and 10 minutes. The results showed that the minimum wear rate of PA6-30%TiO₂ samples at maximum applied load of 20N and run time of 10 minutes conditions. The results obtained are exceptionally promising which concludes the methodology exhibited in this paper has applications for design and manufacture of automotive parts utilizing FDM technology.

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1. Introduction

Additive Manufacturing (AM) is an appropriate name to describe the technologies that build 3D objects by adding layer-upon-layer of material, as opposed to subtractive manufacturing methodologies [1]. The term AM encompasses many technologies including subsets like 3D Printing, Rapid Prototyping (RP), Direct Digital Manufacturing (DDM), layered manufacturing, additive fabrication and selective laser sintering from metals and powders [2]. Fused Deposition Modelling (FDM) has found application in various areas including biomedical, aeronautics or aerospace engineering, which is a growing additive manufacturing (AM) technology [3]. Materials such as ABS, PLA can be used to produce prototypes from FDM a RP system by laying tracks of semi-molten plastic filament on a platform in a layer wise manner from bottom to top [4]. Advancement in materials one of the important factors that allows new design and innovative approaches for AM applications. The improvement of mechanical properties in FDM parts by defining optimal printing parameters configuration, modifying fabrication parameters - treatments and reinforcing parts using composite materials are been focused in recent studies [5-8]. Aluminium alloys has been studied with addition of fibers in polymers in AM and has been concluded with the increase in certain mechanical properties [9]. A composite material is composed of at least two materials, which combine to give properties superior to those of the individual constituents and allows the materials to stay distinct and identifiable. Research on development of composite materials for AM processes has been on-going for more than a decade [10]. The polyamide6 (PA6)/ Nylon6 fibers are wrinkle proof and highly resistant to abrasion and chemicals such as acids and alkalis. They are tough, possessing high tensile strength, as well as elasticity and luster. The fibers can absorb up to 2.4% of water, although this lowers tensile strength. PA6 is a stable polymer and not readily degraded in the ambient environment [11]. TiO_2 is one of the safest and most used inorganic nanostructures for modification of polymeric materials. In addition, Titanium dioxide (TiO_2) is widely used in sunscreen lotions as active broadband sunscreens for high SPF products, a well-known semiconductor photo catalyst and has exclusive properties such as stable, long lasting, secure and broad spectrum [12]. A large amount of irradiance of UV rays on earth's surface due to depletion of ozone layer which creates urgent need to increase the biodegradability of the polymeric material and improve the materials UV protection properties, because [11-12].

It was inferred that incorporation of the exact amount nanoparticles in the polyamide have zero agglomeration and remarkably improved the properties. Surprisingly only a few researches have concentrated on carrying out the experiments on polyamide with TiO_2 nanoparticles. By using TiO_2 the biodegradability of the polyamide material increases. But there is no thorough study on the specified ones. Based on the detailed literature survey, after identifying the research gaps the TiO_2 nanoparticles and PA6 was prepared using FDM and mechanical properties were investigated.

2. Experimental Setup

2.1. Materials and Method

PA6 a semi crystalline polyamide commonly known as nylon 6, has good thermal stability, low dielectric constant, high tensile strength, toughness and high elongation is more commonly used as matrix element. PA6 has a density of 1.14 g/cm^3 , tensile strength which varies from $4.22\text{--}16.88 \text{ kgf/mm}^2$ is high when compared with other polymers. It has a good flexural strength ($274.27\text{--}773.59 \text{ kgf/mm}^2$) with a melting point of 215°C . Recent researches reveal that TiO_2 can be best suited for polymeric material with preview of enhancing bio-degradability and improving biocompatibility. For present study blend of PA6 and TiO_2 (70-30wt. %) in granules form has been selected as target material for investigation because of their all-round material properties and melt processing capabilities [13].

2.2 Extrusion of the material

Before the extrusion process the material is dried at 90°C for minimum of 5hrs. Polyamide 6 (Heraeus Instruments) was compounded with 30 wt. % of titanium-di-oxide (Heraeus Instruments). The dried materials are extruded using by 3Devo-NEXT Filament Extruder. The input parameters were set as barrel temperature 250°C. Speed of extruder is 50 RPM and load on the extruder 15 kg for overall better mechanical properties of PA6/ TiO₂ blend.

2.3 Fusion deposition modelling

Tensile test specimens (ASTM D638 Type 4) were modelled using Solid works. This file is then imported into “slic3r” and the G-codes are generated. The generated g-codes were uploaded in the machine kühlung and kühlung HT500. The specimens were extruded at temperature of 250°C, fill angle 45° and layer height of 0.3 mm and filament diameter of 1.75 mm. The specimens were used to evaluate Tensile strength and wear rate. Figure 1 and Fig 2 illustrates the 3D printer used for printing the specimen

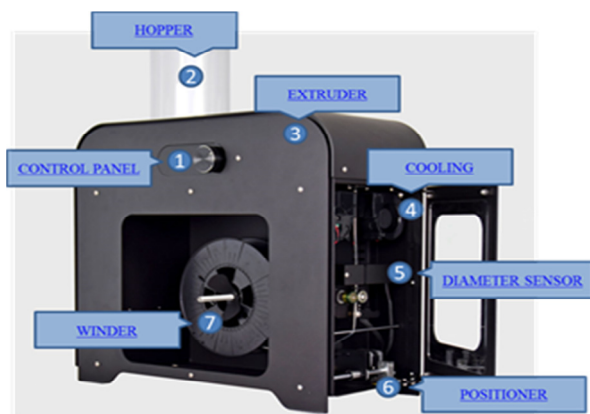


Fig. 1. Filament Extruder



Fig. 2. 3D Printer (FDM)

2.4 Testing of the specimen

The tensile properties of the 3D printed specimens were tested according to ASTM D638 standard in Instron 5584 universal testing machine in Fig 3. The specimens were loaded with a speed of 5 mm/min. The maximum load, elongation at break were noted for pure polymer and three different compositions of polymer and TiO₂. The tested specimens are shown in Fig 4. The tensile strength, Young's modulus and elongation at break has been evaluated using the values and shown in table 1. The 3D printed tensile test specimens of four different combinations were tested and the tested specimens are shown in Fig 4. The specimens in figure 4 are shown in order with respect to the composition of PA6 and TiO₂ with virgin PA6 at the top to a mixture of PA6+30% TiO₂ at the bottom.

Table 1. Mechanical properties of extruded material

Sample	Tensile strength (Mpa)	Young modulus (Mpa)	Elongation at break (%)
Virgin PA6	51.6	452	35.6%
PA6+10%TiO ₂	57.8	494	25.60%
PA6+20%TiO ₂	59.5	513	16.14%
PA6+30%TiO ₂	62.6	532	13.35%



Fig. 3. Instron 5584 universal testing machine



Fig. 4. Tensile tested specimens

2.5. Tribometer testing

The tribological behaviours of the 3D printed test samples were investigated in pin on disc wear tribometer. The prepared specimens were examined by using Ducom (TR-20LE-M108) tribometer machine to vary the influenced process parameters such as applied load 5,10,15,20 N and run time 5mins, 10mins respectively, also other process parameters were kept uninterrupted such as sliding speed 1.256 m/s, track diameter 240 mm, disc speed 100 rpm. The tribological behaviours such as wear rate, friction force, the coefficient of friction are taken for four test samples and the values are noted for further analysis as in Table 2. The wear test specimens of four different compositions of PA6 and TiO₂ are shown in figure 6. Each specimen of different compositions were tested under different load conditions and different run time and as shown in below Fig 5 and Fig 6.



Fig. 5. Pin on disk tribometer

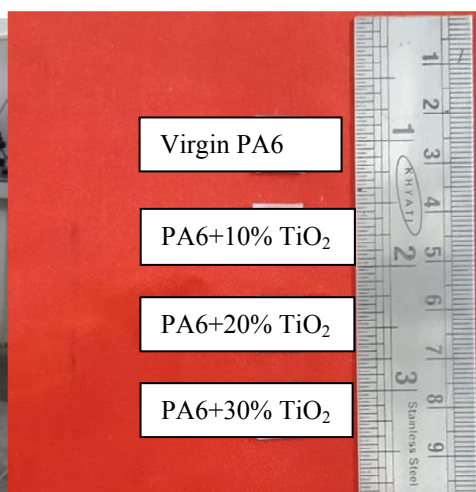


Fig. 6. Wear test samples

Table 2. Wear rate of FDM printed material

Material	Load (N)	Load at 100 rpm (5mins)	Load at 100 rpm (10mins)
		Wear rate (mm ³ /Nm)	Wear rate (mm ³ /Nm)
PA6	5	1.75	1.65
	10	3.2	3.1
	15	4.5	4.2
	20	5.2	4.82
PA6+10%TiO ₂	5	1.35	1.25
	10	2.4	2.1
	15	2.85	2.55
	20	3.64	2.8
PA6+20% TiO ₂	5	1.1	1
	10	1.9	1.6
	15	2.325	1.8
	20	2.84	2.0
PA6+30% TiO ₂	5	0.95	0.8
	10	1.6	1.2
	15	1.8	1.35
	20	2.0	1.6

3. Result and Discussion

3.1 Tensile strength

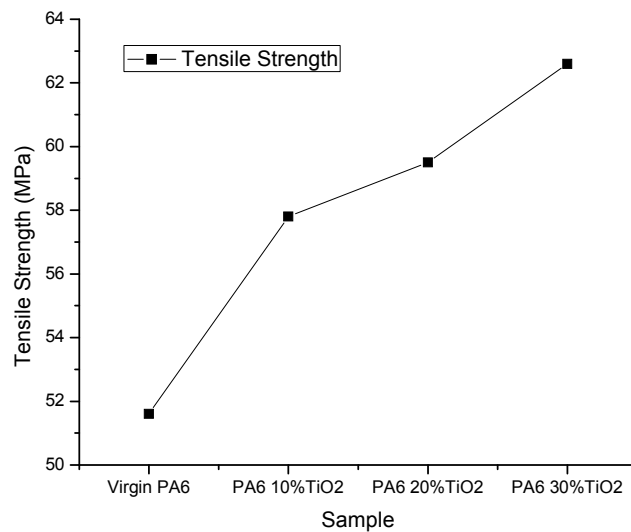


Fig. 7. Tensile strength

The Tensile strength of various mixtures of PA6 TiO₂ nanocomposites is appeared in Fig 7. It becomes evident that there is an improvement in tensile strength of virgin PA6 due to reinforcement addition. The tensile strength of PA6 & TiO₂ composites were improved by 12% and 16% with TiO₂ 10% and 20% when compared with Virgin PA6. The strength of PA6 & TiO₂ steadily increased up to 30% of TiO₂. It was noted that PA6 & 30% of TiO₂ had a tensile strength of 62.6Mpa which was 22% higher than that of virgin PA6. Incorporation of TiO₂ with virgin polymer blend resulted in improvement in tensile strength. Further incorporation of TiO₂ results in gradual improvement in tensile strength of neat polymer blend. This demonstrate that different compositions of PA6- TiO₂ offers higher resistance to deformation because of TiO₂ nanoparticles content, consequently the stiffness of the PA6-TiO₂ composite under axial loading improves.

3.2 Young's modulus

Young's modulus is defined as the ratio of stress below the proportional limit to the corresponding strain. It is a measure of the rigidity or stiffness of a material. In terms of the stress-strain curve, Young's modulus is the slope of the stress-strain curve in the range of linear proportionality of stress to strain. Young's modulus has been calculated from obtained results and shown in Fig 8. As we discussed earlier, incorporation of TiO₂ with unadulterated PA6 blend enhanced the tensile strength of the pure polymer blend. The elasticity influences the Young's modulus of the PA6- TiO₂ composites. PA6+10%wt TiO₂ reveals an increased Young's modulus of 494 MPa which is about 10% greater than that of virgin PA6. Similarly PA6+20%wt TiO₂ and PA6+30%wt TiO₂ composites has the Young's modulus of 513 MPa and 532 MPa respectively which is about 14% and 18% higher than that of neat polymer blend. Blending of TiO₂ with PA6 results in impulsive change of Young's modulus. The Young's modulus of the polymer blend improved gradually with increase of TiO₂.

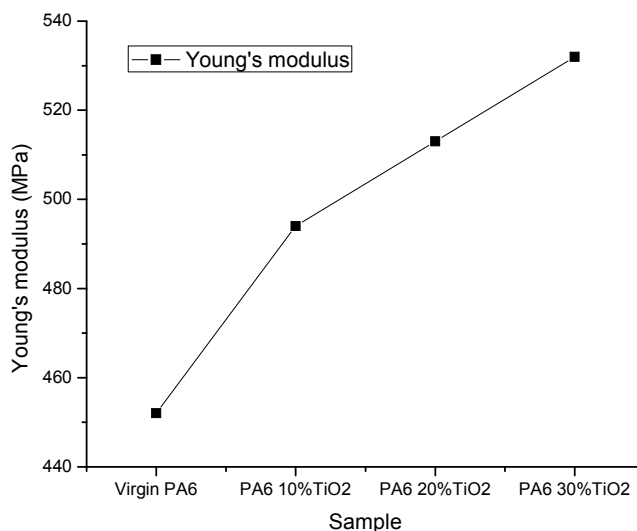


Fig. 8. Young's modulus

3.3 Percentage of elongation at break

The elongation at break of the different compositions of the PA6- TiO₂ blends exhibits in Fig 9. Fracture strain is the proportion of changed length to the original length after breakage of the specimen. The elongation of virgin PA6 is compared with PA6 and TiO₂. PA6+10%wt TiO₂ has the elongation up to 25.6% which is about 39% less than that of pure PA6 blend. PA6+20%wt TiO₂ composite and PA6+30%wt TiO₂ has the elongation up to 16.14% and

13.35% respectively which is about 55% and 62% less than that of neat polymer blend. This shows that fusion of TiO_2 with pure polymer has reduced the elasticity of the pure polymer blend drastically. Addition of further filler results in drastic decrease in the elasticity of the pure polymer blend.

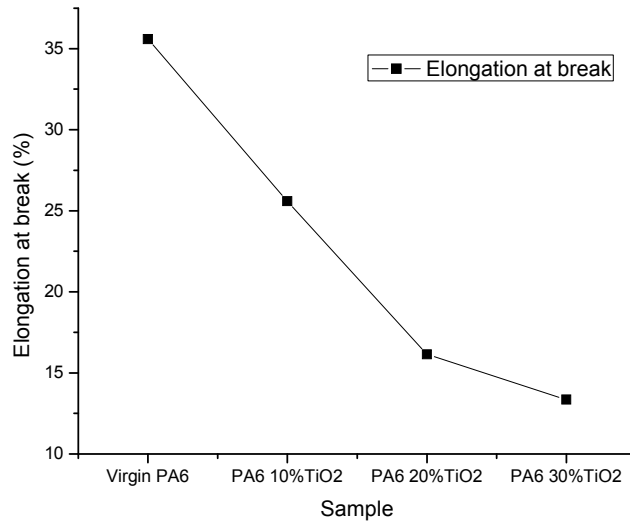


Fig. 9. Elongation at break

3.4. Wear rate

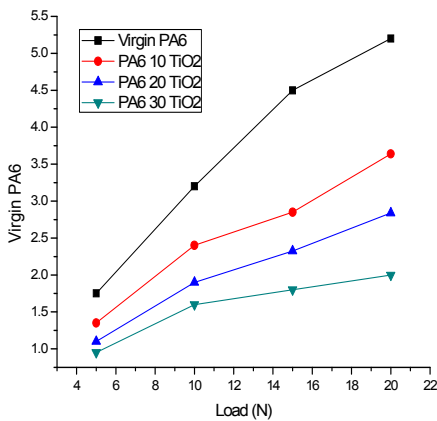


Fig. 10. Wear rate for 5mins

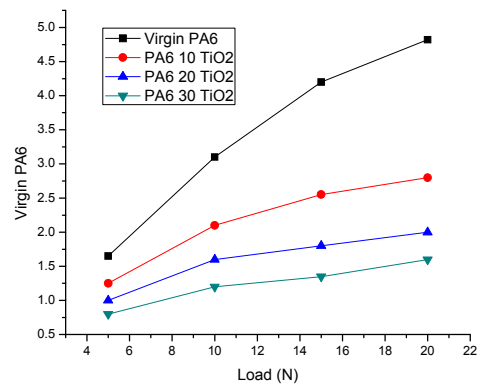


Fig. 11. Wear rate for 10mins

Wear is the loss of material from the surface of a solid body due to mechanical action, i.e. contact and relative movement of a solid, fluid or gaseous counter body. The resistance to wear for polymers enhances with the use of reinforcement filler materials. By adding TiO_2 in a varying weight proportion, the experimental wear rate of the respective test samples were made significant. From the Fig. 10 and Fig. 11, the wear rate decreases with increase in weight percentage of TiO_2 due to enhancement of hardness and fatigue strength. The addition of TiO_2 increases the

abrasion and sliding resistance due to its high thermal stability. The wear rate is observed to be peak for all the test samples due to the less adhesion between the surfaces at lower load conditions. At higher loads, the adhesion between the surfaces gets strengthened and makes the material to wear more in sliding condition. For PA6 the value increases steadily with the increase in applied load and for other composites the wear rate increase gradually from 5 to 15 N, after that, it becomes almost stable for load of 15 to 20 N owing to sliding resistance. The reinforcement of TiO_2 in PA6 matrix prevents the transition from mild to severe wear at high load condition. The loads are applied for specific time period of 5mins and 10mins. The fatigue stress increases with the rise in the time period which can be seen from the test. The wear test reveals that there is heat dissipation from the contact surface of pin and disk due to friction which confirms the impale of [14]. This further decreases the wear rate of material. The PA6+10% TiO_2 and PA6+20% TiO_2 composites have 41% and 58.5% lesser value than the virgin PA6 for 20N load and 10mins time period. The wear rate of PA6+30% TiO_2 is 66.8 % lesser than the virgin PA6 for the same conditions, which is the eminent value for the taken samples. The experiment reveals that PA6+30% TiO_2 have better wear resistance than the virgin PA6 and other composites.

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4. Conclusion

It has been derived from the present research, the printed samples of PA6 & TiO_2 in FDM process has been contributed as takes after.

1. Higher properties were attained due to non-agglomeration of TiO_2 at 30wt% of TiO_2 in the matrix of printed samples.
2. After incorporation of 10wt% of TiO_2 , tensile strength and Young's modulus of the polymer matrix were increased by 10% and 12% respectively. Whereas elongation is decreased by 41%.
3. Incorporation of 20wt% of TiO_2 has increased the tensile strength and Young's modulus of the virgin polymer by 16% and 14% respectively. However, elongation is reduced by 58.5%.
4. Incorporation of 30wt% of TiO_2 increased the tensile strength and Young's modulus of the pure polymer by 22% and 18% respectively. Though elongation is decreased by 66.8%.
5. Fusion of TiO_2 in all three composition (10wt%, 20wt% and 30wt %) has positively affected the tensile strength of the pure polyamide.
6. The addition of TiO_2 increases the biodegradability, abrasion; sliding resistance due to high thermal stability and so PA6+30% TiO_2 has the lower wear rate for 20N load at 10mins.
7. As the percentage of TiO_2 increases, the agglomeration of TiO_2 nanoparticles increases which in reduced the tensile strength; an optimum level of 30% TiO_2 was incorporated.
8. This composite can be suitable for mechanical and tribological related applications in automotive industry; also it is owing to an extended life span due to its properties and also quick bio degradability of this composite

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